

Psychedelics as a Dual Therapy: Exploring Their Potential in Treating Addictive Disorders and Inflammation

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BACKGROUND & OBJECTIVE

Alcohol use disorder (AUD), frequently **comorbid with depression**, has been associated with **inflammation** and disturbances in the **gut-immune-brain axis** (1,2,3). Psychedelics show **therapeutic promise in both conditions** (4,5,6), but their potential immunomodulatory effects remain insufficiently characterized in humans (7).

Objective

To systematically review human and human-derived evidence on the **inflammatory and immunomodulatory effects of psychedelics** and **explore their relevance to inflammatory phenotypes in psychiatric and addictive disorders**.

METHODS

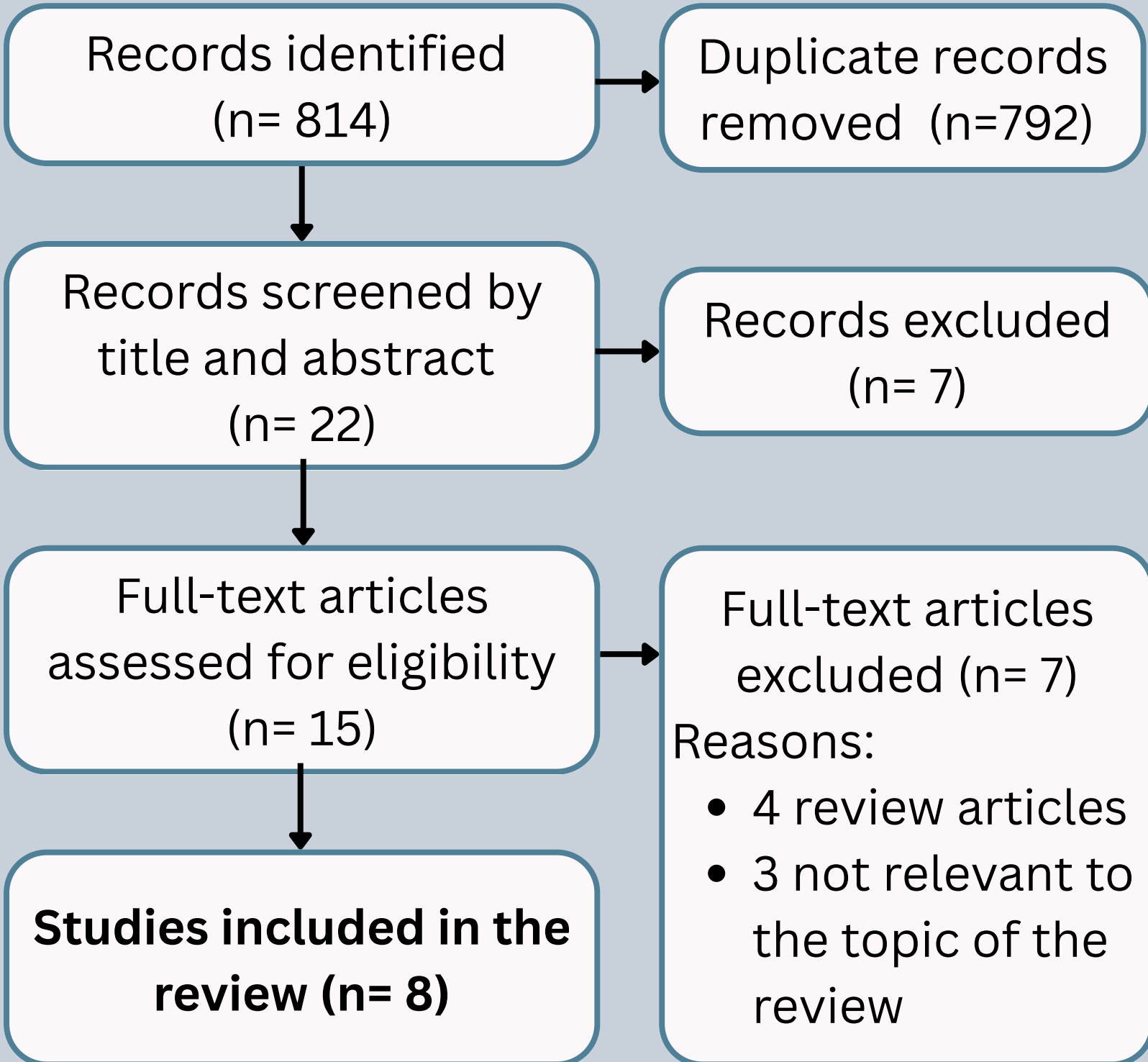
Systematic review following PRISMA guidelines
Search through **January 1, 2026**

Databases: PubMed, Google Scholar, Web of Science, Microsoft Academic, Crossref, OpenAlex, and Semantic Scholar

Included: human clinical studies and human-derived in vitro studies assessing inflammatory/immune outcomes following psychedelic exposure

Excluded: reviews, animal studies and non-human cellular models

Study selection (PRISMA)



RESULTS

8 STUDIES INCLUDED | Clinical studies (n = 4)
Human-derived in vitro studies (n = 4)

Psychedelics showed **selective** and **context-dependent immunomodulatory effects**, with **reductions in pro-inflammatory signaling** across clinical and cellular studies.

Compound	Type of study	Selected findings	References
Psilocybin/psilocin	Human clinical + in vitro	↓ TNF-α (acute) (8) ↓ IL-6 and CRP (day 7) (8) In vitro: ↓ ROS and modulation of microglial functions (9)	(8) Mason et al., 2023 (9) Wiens et al., 2024 (10) Burmester et al., 2023
Psilocybin-containing mushroom extracts	In vitro (human-derived)	↓ COX-2 (11) ↓ TNF-α, IL-1β and IL-6 (11)	(11) Nkadameng et al., 2021
Ayahuasca	Human clinical	↓ CRP (48 hours) in treatment-resistant depression (12) Reduction in CRP associated with clinical improvement (12)	(12) Galvão-Coelho et al., 2020 (13) de Sousa et al., 2022
N,N-DMT/5-MeO-DMT	Human clinical + in vitro	Clinical: ↓ salivary IL-6 immediately post-session (14) In vitro: ↓ TNF-α, IL-1β, IL-6 and IL-8; ↑ IL-10; reduced Th1/Th17 responses (15)	(14) Uthaug et al., 2020 (15) Szabo et al., 2014

PROPOSED IMMUNOMODULATORY PATHWAYS

The available evidence suggests **two potentially complementary pathways** underlying the **immunomodulatory effects** of psychedelics

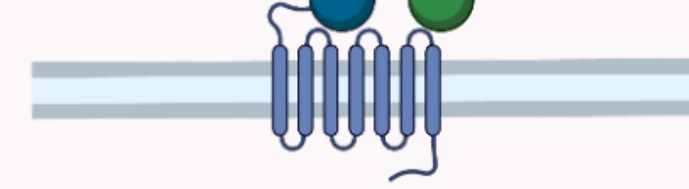


DMT/5-MeO-DMT compounds

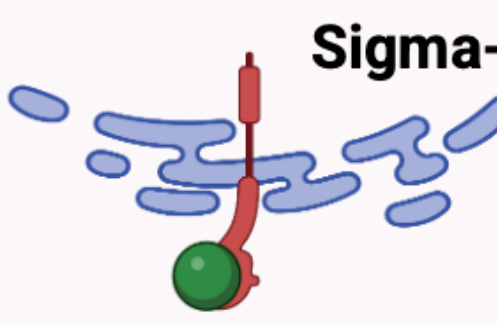


Psilocybin/Psilocin

5-HT_{2A}AR



G protein-coupled serotonin receptor located at the plasma membrane.



Sigma-1 (S1R)

Intracellular chaperone protein located primarily at the ER-mitochondria interface.

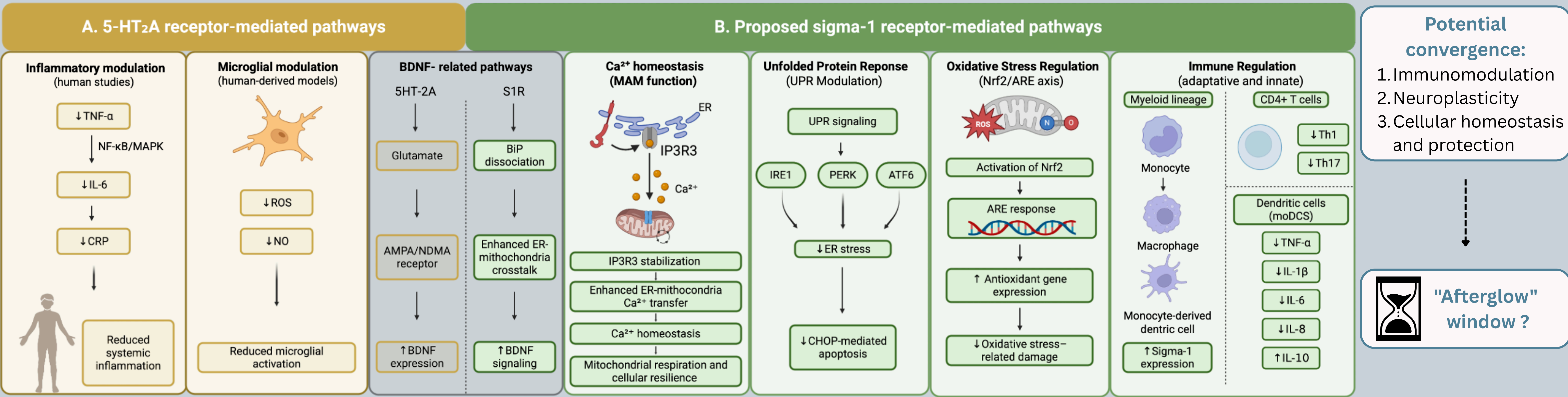


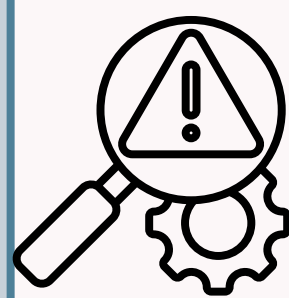
Figure 1: Conceptual model integrating review findings and broader mechanistic literature. Causal pathways and their contribution to clinical outcomes remain unconfirmed.

Clinical implications



- **Baseline inflammation warrants evaluation** as a predictor of **clinical response** and immunomodulatory effects following psychedelic administration.
- Findings provide a **rationale for investigating immunomodulation in AUD** and psychiatric disorders.

Limitations



- Eight studies with **heterogeneous designs and biomarker timing**.
- Peripheral biomarkers do not establish central neuroimmune effects.
- **No direct evidence in AUD** or proof that immune changes mediate clinical benefit.

Conclusions

Psychedelics may **modulate immune responses**, but the **therapeutic significance of these effects remains uncertain**. Future trials should link biomarker changes to clinical outcomes.

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References: (1) Vigo et al., 2025; (2) Leclercq et al., 2021; (3) Niu et al., 2026; (4) Luquiens et al., 2025; (5) Bogenschutz et al., 2022; (6) Metaxa and Clarke, 2024; (7) Low et al., 2025;

